Kubernetes - First Steps

Preconditions:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl version
Client Version: v1.32.5+k3s1
Kustomize Version: v5.5.0
Server Version: v1.32.5+k3s1
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ k3s kubectl get nodes
NAME               STATUS   ROLES                  AGE   VERSION
ip-172-31-18-113   Ready    control-plane,master   24h   v1.32.5+k3s1
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Pod Deployment:

Create a single nginx Pod

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl run nginx --image=nginx
pod/nginx created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Verify

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get pod
NAME    READY   STATUS    RESTARTS   AGE
nginx   1/1     Running   0          28s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Configuration files:

Create wordpress.yml file

```yaml
# wordpress.yml U
# datascientest-data-engineering > kubernetes > wordpress.yml
apiVersion: v1
kind: Pod
metadata:
  name: wordpress
spec:
  containers:
  - name: wordpress
    image: wordpress
    ports:
    - containerPort: 80
```

Apply configuration to create the resource

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f wordpress.yml
pod/wordpress created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Check the list of existing Pods

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get pod
NAME        READY   STATUS    RESTARTS   AGE
nginx       1/1     Running   0          8m20s
wordpress   1/1     Running   0          83s
```

Delete both pods

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl delete pod nginx wordpress
pod "nginx" deleted
pod "wordpress" deleted
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Observe self-healing

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get pod
No resources found in default namespace.
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Deployments:

10-replica nginx deployment

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl create deployment nginx-deployment --image=nginx --replicas=10
deployment.apps/nginx-deployment created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Retrieve all available resources

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get all
NAME                                    READY   STATUS    RESTARTS   AGE
pod/nginx-deployment-6cfb98644c-6g9fp   1/1     Running   0          71s
pod/nginx-deployment-6cfb98644c-7b7lw   1/1     Running   0          71s
pod/nginx-deployment-6cfb98644c-dl9s8   1/1     Running   0          71s
pod/nginx-deployment-6cfb98644c-dzp7w   1/1     Running   0          71s
pod/nginx-deployment-6cfb98644c-g7ndn   1/1     Running   0          71s
pod/nginx-deployment-6cfb98644c-ks4g6   1/1     Running   0          71s
pod/nginx-deployment-6cfb98644c-n6l29   1/1     Running   0          71s
pod/nginx-deployment-6cfb98644c-t2r47   1/1     Running   0          71s
pod/nginx-deployment-6cfb98644c-w7l46   1/1     Running   0          71s
pod/nginx-deployment-6cfb98644c-z5vwv   1/1     Running   0          71s

NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.43.0.1    <none>        443/TCP   24h

NAME                               READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/nginx-deployment   10/10   10           10          71s

NAME                                          DESIRED   CURRENT   READY   AGE
replicaset.apps/nginx-deployment-6cfb98644c   10        10        10      71s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Clean up

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl delete deployment nginx-deployment
deployment.apps "nginx-deployment" deleted
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Create deployment.yml

! deployment.yml U ×

datascientest-data-engineering > kubernetes > ! deployment.yml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: web
spec:
  selector:
    matchLabels:
      app: web
  replicas: 10
  strategy:
    type: RollingUpdate
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
        - name: nginx
          image: nginx
          ports:
            - containerPort: 80
```

Apply this configuration to create Deployment resource

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f deployment.yml
deployment.apps/nginx-deployment created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Observe list of resources within a kubernetes cluster

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl api-resources
NAME                                SHORTNAMES   APIVERSION                          NAMESPACED   KIND
bindings                                         v1                                  true         Binding
componentstatuses                   cs           v1                                  false        ComponentStatus
configmaps                          cm           v1                                  true         ConfigMap
endpoints                           ep           v1                                  true         Endpoints
events                              ev           v1                                  true         Event
limitranges                         limits       v1                                  true         LimitRange
namespaces                          ns           v1                                  false        Namespace
nodes                               no           v1                                  false        Node
persistentvolumeclaims              pvc          v1                                  true         PersistentVolumeClaim
persistentvolumes                   pv           v1                                  false        PersistentVolume
pods                                po           v1                                  true         Pod
podtemplates                                     v1                                  true         PodTemplate
replicationcontrollers              rc           v1                                  true         ReplicationController
resourcequotas                      quota        v1                                  true         ResourceQuota
secrets                                          v1                                  true         Secret
serviceaccounts                     sa           v1                                  true         ServiceAccount
services                            svc          v1                                  true         Service
mutatingwebhookconfigurations                    admissionregistration.k8s.io/v1     false        MutatingWebhookConfiguration
validatingadmissionpolicies                      admissionregistration.k8s.io/v1     false        ValidatingAdmissionPolicy
validatingadmissionpolicybindings                admissionregistration.k8s.io/v1     false        ValidatingAdmissionPolicyBinding
validatingwebhookconfigurations                  admissionregistration.k8s.io/v1     false        ValidatingWebhookConfiguration
customresourcedefinitions           crd,crds     apiextensions.k8s.io/v1             false        CustomResourceDefinition
apiservices                                      apiregistration.k8s.io/v1           false        APIService
controllerrevisions                              apps/v1                             true         ControllerRevision
daemonsets                          ds           apps/v1                             true         DaemonSet
deployments                         deploy       apps/v1                             true         Deployment
replicasets                         rs           apps/v1                             true         ReplicaSet
statefulsets                        sts          apps/v1                             true         StatefulSet
selfsubjectreviews                               authentication.k8s.io/v1            false        SelfSubjectReview
tokenreviews                                     authentication.k8s.io/v1            false        TokenReview
localsubjectaccessreviews                        authorization.k8s.io/v1             true         LocalSubjectAccessReview
selfsubjectaccessreviews                         authorization.k8s.io/v1             false        SelfSubjectAccessReview
selfsubjectrulesreviews                          authorization.k8s.io/v1             false        SelfSubjectRulesReview
subjectaccessreviews                             authorization.k8s.io/v1             false        SubjectAccessReview
horizontalpodautoscalers            hpa          autoscaling/v2                      true         HorizontalPodAutoscaler
cronjobs                            cj           batch/v1                            true         CronJob
jobs                                             batch/v1                            true         Job
certificatesigningrequests          csr          certificates.k8s.io/v1              false        CertificateSigningRequest
leases                                           coordination.k8s.io/v1              true         Lease
endpointslices                                   discovery.k8s.io/v1                 true         EndpointSlice
events                              ev           events.k8s.io/v1                    true         Event
flowschemas                                      flowcontrol.apiserver.k8s.io/v1     false        FlowSchema
prioritylevelconfigurations                      flowcontrol.apiserver.k8s.io/v1     false        PriorityLevelConfiguration
gatewayclasses                      gc           gateway.networking.k8s.io/v1        false        GatewayClass
gateways                            gtw          gateway.networking.k8s.io/v1        true         Gateway
grpcroutes                                       gateway.networking.k8s.io/v1        true         GRPCRoute
httproutes                                       gateway.networking.k8s.io/v1        true         HTTPRoute
referencegrants                     refgrant     gateway.networking.k8s.io/v1beta1   true         ReferenceGrant
helmchartconfigs                                 helm.cattle.io/v1                   true         HelmChartConfig
helmcharts                                       helm.cattle.io/v1                   true         HelmChart
accesscontrolpolicies                            hub.traefik.io/v1alpha1             false        AccessControlPolicy
aiservices                                       hub.traefik.io/v1alpha1             true         AIService
apiaccesses                                      hub.traefik.io/v1alpha1             true         APIAccess
apibundles                                       hub.traefik.io/v1alpha1             true         APIBundle
apicatalogitems                                  hub.traefik.io/v1alpha1             true         APICatalogItem
apiplans                                         hub.traefik.io/v1alpha1             true         APIPlan
apiportals                                       hub.traefik.io/v1alpha1             true         APIPortal
apiratelimits                                    hub.traefik.io/v1alpha1             true         APIRateLimit
apis                                             hub.traefik.io/v1alpha1             true         API
apiversions                                      hub.traefik.io/v1alpha1             true         APIVersion
managedsubscriptions                             hub.traefik.io/v1alpha1             true         ManagedSubscription
addons                                           k3s.cattle.io/v1                    true         Addon
etcdsnapshotfiles                                k3s.cattle.io/v1                    false        ETCDSnapshotFile
nodes                                            metrics.k8s.io/v1beta1              false        NodeMetrics
```

```
pods                                             metrics.k8s.io/v1beta1              true         PodMetrics
ingressclasses                                   networking.k8s.io/v1                false        IngressClass
ingresses                           ing          networking.k8s.io/v1                true         Ingress
networkpolicies                     netpol       networking.k8s.io/v1                true         NetworkPolicy
runtimeclasses                                   node.k8s.io/v1                      false        RuntimeClass
poddisruptionbudgets                pdb          policy/v1                           true         PodDisruptionBudget
clusterrolebindings                              rbac.authorization.k8s.io/v1        false        ClusterRoleBinding
clusterroles                                     rbac.authorization.k8s.io/v1        false        ClusterRole
rolebindings                                     rbac.authorization.k8s.io/v1        true         RoleBinding
roles                                            rbac.authorization.k8s.io/v1        true         Role
priorityclasses                     pc           scheduling.k8s.io/v1                false        PriorityClass
csidrivers                                       storage.k8s.io/v1                   false        CSIDriver
csinodes                                         storage.k8s.io/v1                   false        CSINode
csistoragecapacities                             storage.k8s.io/v1                   true         CSIStorageCapacity
storageclasses                      sc           storage.k8s.io/v1                   false        StorageClass
volumeattachments                                storage.k8s.io/v1                   false        VolumeAttachment
ingressroutes                                    traefik.io/v1alpha1                 true         IngressRoute
ingressroutetcps                                 traefik.io/v1alpha1                 true         IngressRouteTCP
ingressrouteudps                                 traefik.io/v1alpha1                 true         IngressRouteUDP
middlewares                                      traefik.io/v1alpha1                 true         Middleware
middlewaretcps                                   traefik.io/v1alpha1                 true         MiddlewareTCP
serverstransports                                traefik.io/v1alpha1                 true         ServersTransport
serverstransporttcps                             traefik.io/v1alpha1                 true         ServersTransportTCP
tlsoptions                                       traefik.io/v1alpha1                 true         TLSOption
tlsstores                                        traefik.io/v1alpha1                 true         TLSStore
traefikservices                                  traefik.io/v1alpha1                 true         TraefikService
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Verify all pods

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get all
NAME                                    READY   STATUS    RESTARTS   AGE
pod/nginx-deployment-6476b5cb9d-bvvtt   1/1     Running   0          7m12s
pod/nginx-deployment-6476b5cb9d-djvp7   1/1     Running   0          7m12s
pod/nginx-deployment-6476b5cb9d-fq8rb   1/1     Running   0          7m12s
pod/nginx-deployment-6476b5cb9d-fvhhl   1/1     Running   0          7m12s
pod/nginx-deployment-6476b5cb9d-h7pg5   1/1     Running   0          7m12s
pod/nginx-deployment-6476b5cb9d-t5f7f   1/1     Running   0          7m12s
pod/nginx-deployment-6476b5cb9d-tqhjz   1/1     Running   0          7m12s
pod/nginx-deployment-6476b5cb9d-v8xfx   1/1     Running   0          7m12s
pod/nginx-deployment-6476b5cb9d-x9jp9   1/1     Running   0          7m12s
pod/nginx-deployment-6476b5cb9d-xxbtd   1/1     Running   0          7m12s

NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.43.0.1    <none>        443/TCP   24h

NAME                               READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/nginx-deployment   10/10   10           10          7m12s

NAME                                          DESIRED   CURRENT   READY   AGE
replicaset.apps/nginx-deployment-6476b5cb9d   10        10        10      7m12s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Delete a pod

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl delete Pod nginx-deployment-6476b5cb9d-bvvtt
pod "nginx-deployment-6476b5cb9d-bvvtt" deleted
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Observe new Pod spin up

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get pod
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6476b5cb9d-2jtr7   1/1     Running   0          40s
nginx-deployment-6476b5cb9d-djvp7   1/1     Running   0          15m
nginx-deployment-6476b5cb9d-fq8rb   1/1     Running   0          15m
nginx-deployment-6476b5cb9d-fvhhl   1/1     Running   0          15m
nginx-deployment-6476b5cb9d-h7pg5   1/1     Running   0          15m
nginx-deployment-6476b5cb9d-t5f7f   1/1     Running   0          15m
nginx-deployment-6476b5cb9d-tqhjz   1/1     Running   0          15m
nginx-deployment-6476b5cb9d-v8xfx   1/1     Running   0          15m
nginx-deployment-6476b5cb9d-x9jp9   1/1     Running   0          15m
nginx-deployment-6476b5cb9d-xxbtd   1/1     Running   0          15m
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Services:

Expose nginx-deployment inside our cluster

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl expose deploy nginx-deployment --port=80 --type=ClusterIP
service/nginx-deployment exposed
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Display list of services

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get service
NAME               TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
kubernetes         ClusterIP   10.43.0.1       <none>        443/TCP   25h
nginx-deployment   ClusterIP   10.43.229.119   <none>        80/TCP    56s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Check access to Pods managed by the nginx deployment / test from inside cluster

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl run curl --image=curlimages/curl -i --tty -- sh
If you don't see a command prompt, try pressing enter.
~ $ curl 10.43.229.119
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
~ $ exit
Session ended, resume using 'kubectl attach curl -c curl -i -t' command when the pod is running
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Create service-clusterip.yml

```yaml
service-clusterip.yml U
datascientest-data-engineering > kube
apiVersion: v1
kind: Service
metadata:
  name: nginx-cip
spec:
  selector:
    app: web
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
```

Apply and check

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f service-clusterip.yml
service/nginx-cip created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get svc
NAME               TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
kubernetes         ClusterIP   10.43.0.1       <none>        443/TCP   25h
nginx-cip          ClusterIP   10.43.117.250   <none>        80/TCP    8s
nginx-deployment   ClusterIP   10.43.229.119   <none>        80/TCP    12m
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get pod --show-labels
NAME                                READY   STATUS    RESTARTS       AGE    LABELS
curl                                1/1     Running   1 (5m42s ago)  8m4s   run=curl
nginx-deployment-6476b5cb9d-2jtr7   1/1     Running   0              21m    app=web,pod-template-hash=6476b5cb9d
nginx-deployment-6476b5cb9d-djvp7   1/1     Running   0              36m    app=web,pod-template-hash=6476b5cb9d
nginx-deployment-6476b5cb9d-fq8rb   1/1     Running   0              36m    app=web,pod-template-hash=6476b5cb9d
nginx-deployment-6476b5cb9d-fvhhl   1/1     Running   0              36m    app=web,pod-template-hash=6476b5cb9d
nginx-deployment-6476b5cb9d-h7pg5   1/1     Running   0              36m    app=web,pod-template-hash=6476b5cb9d
nginx-deployment-6476b5cb9d-t5f7f   1/1     Running   0              36m    app=web,pod-template-hash=6476b5cb9d
nginx-deployment-6476b5cb9d-tqhjz   1/1     Running   0              36m    app=web,pod-template-hash=6476b5cb9d
nginx-deployment-6476b5cb9d-v8xfx   1/1     Running   0              36m    app=web,pod-template-hash=6476b5cb9d
nginx-deployment-6476b5cb9d-x9jp9   1/1     Running   0              36m    app=web,pod-template-hash=6476b5cb9d
nginx-deployment-6476b5cb9d-xxbtd   1/1     Running   0              36m    app=web,pod-template-hash=6476b5cb9d
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Delete clusterIP service

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl delete svc nginx-deployment
service "nginx-deployment" deleted
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Create NodePort service type

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl expose deployment nginx-deployment --port=80 --type=NodePort
service/nginx-deployment exposed
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Check the dynamic port assigned to connect to our service

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get service
NAME               TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kubernetes         ClusterIP   10.43.0.1       <none>        443/TCP        25h
nginx-cip          ClusterIP   10.43.117.250   <none>        80/TCP         11m
nginx-deployment   NodePort    10.43.25.26     <none>        80:31855/TCP   46s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

With VM IP 176.34.128.215 and port 31855, visit browser by opening URL http://176.34.128.215:31855/

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org. Commercial support is available at nginx.com.

Thank you for using nginx.

Create serviceNodePort.yml

serviceNodePort.yml U

datascientest-data-engineering > kuber

```yaml
apiVersion: v1
kind: Service
metadata:
  name: monservicenodeport
spec:
  type: NodePort
  selector:
    app: web
  ports:
    - name: http
      protocol: TCP
      port: 80
      targetPort: 80
      nodePort: 30000
```

Check list of services present

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f serviceNodePort.yml
service/monservicenodeport created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get endpoints monservicenodeport
NAME                 ENDPOINTS                                              AGE
monservicenodeport   10.42.0.36:80,10.42.0.38:80,10.42.0.39:80 + 7 more...  5s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get svc
NAME                 TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kubernetes           ClusterIP   10.43.0.1       <none>        443/TCP        25h
monservicenodeport   NodePort    10.43.89.24     <none>        80:30000/TCP   66s
nginx-cip            ClusterIP   10.43.117.250   <none>        80/TCP         36m
nginx-deployment     NodePort    10.43.25.26     <none>        80:31855/TCP   25m
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Browse to URL http://176.34.128.215:30000/

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

LoadBalancer:
Install MetalLB's manifests

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f https://raw.githubusercontent.com/metallb/metallb/v0.13.10/config/manifests/metallb-n
ative.yaml
namespace/metallb-system created
customresourcedefinition.apiextensions.k8s.io/addresspools.metallb.io created
customresourcedefinition.apiextensions.k8s.io/bfdprofiles.metallb.io created
customresourcedefinition.apiextensions.k8s.io/bgpadvertisements.metallb.io created
customresourcedefinition.apiextensions.k8s.io/bgppeers.metallb.io created
customresourcedefinition.apiextensions.k8s.io/communities.metallb.io created
customresourcedefinition.apiextensions.k8s.io/ipaddresspools.metallb.io created
customresourcedefinition.apiextensions.k8s.io/l2advertisements.metallb.io created
serviceaccount/controller created
serviceaccount/speaker created
role.rbac.authorization.k8s.io/controller created
role.rbac.authorization.k8s.io/pod-lister created
clusterrole.rbac.authorization.k8s.io/metallb-system:controller created
clusterrole.rbac.authorization.k8s.io/metallb-system:speaker created
rolebinding.rbac.authorization.k8s.io/controller created
rolebinding.rbac.authorization.k8s.io/pod-lister created
clusterrolebinding.rbac.authorization.k8s.io/metallb-system:controller created
clusterrolebinding.rbac.authorization.k8s.io/metallb-system:speaker created
configmap/metallb-excludel2 created
secret/webhook-server-cert created
service/webhook-service created
deployment.apps/controller created
daemonset.apps/speaker created
validatingwebhookconfiguration.admissionregistration.k8s.io/metallb-webhook-configuration created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Verify MetalLB controller is running

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get pods -n metallb-system
NAME                          READY   STATUS    RESTARTS   AGE
controller-5c8796d8b6-bcvg9   1/1     Running   0          63m
speaker-wm4tr                 1/1     Running   0          63m
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Create metallb-pool.yml

metallb-pool.yml U ×

datascientest-data-engineering > kubernetes > !

```yaml
apiVersion: metallb.io/v1beta1
kind: IPAddressPool
metadata:
  namespace: metallb-system
  name: default-pool
spec:
  addresses:
    - 176.34.128.200-176.34.128.210
---
apiVersion: metallb.io/v1beta1
kind: L2Advertisement
metadata:
  namespace: metallb-system
  name: default-l2
spec: {}
```

Apply LoadBalancer service

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f metallb-pool.yml
ipaddresspool.metallb.io/default-pool created
l2advertisement.metallb.io/default-l2 created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Create serviceLoadBalancer.yml

```
! serviceLoadBalancer.yml U ×
datascientest-data-engineering >
apiVersion: v1
kind: Service
metadata:
  name: nginx-lb
spec:
  type: LoadBalancer
  selector:
    app: web
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
```

Apply serviceLoadBalancer.yml file

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f serviceLoadBalancer.yml
service/nginx-lb created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Watch for an External IP

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get svc nginx-lb --watch
NAME       TYPE           CLUSTER-IP      EXTERNAL-IP      PORT(S)        AGE
nginx-lb   LoadBalancer   10.43.159.253   176.34.128.202   80:32116/TCP   46s
```

Browse to URL http://176.34.128.202/ to view the Nginx welcome page

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Replicaset Kubernetes:

Check list of replicasets present

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get replicaset
NAME                          DESIRED   CURRENT   READY   AGE
nginx-deployment-6476b5cb9d   10        10        10      3h35m
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Change number of replicas in nginx-deployment from 10 to 1 Pod (for scenario where application demand decreases)

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl scale deploy nginx-deployment --replic
as=1
deployment.apps/nginx-deployment scaled
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Check Replicaset status

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get rs
NAME                          DESIRED   CURRENT   READY   AGE
nginx-deployment-6476b5cb9d   1         1         1       3h37m
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Track number of Pods

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get pod
NAME                                READY   STATUS    RESTARTS       AGE
curl                                1/1     Running   1 (3h6m ago)   3h8m
nginx-deployment-6476b5cb9d-fvhhl   1/1     Running   0              3h37m
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Create replicaset.yml

```yaml
kind: ReplicaSet
metadata:
    name: httpd-replicaset
spec:
    replicas: 3
    selector:
      matchLabels:
        app: httpd
    template:
      metadata:
        labels:
          app: httpd
          env: dev
      spec:
        containers:
        - name: httpd
          image: httpd
```

Apply config to our cluster

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl create -f replicaset.yml
replicaset.apps/httpd-replicaset created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Display list of Replicaset currently running

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get replicaset
NAME                          DESIRED   CURRENT   READY   AGE
httpd-replicaset              3         3         3       29s
nginx-deployment-6476b5cb9d   1         1         1       3h40m
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Provide more details about Replicaset and events created using describe command

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl describe replicaset httpd-replicaset
Name:         httpd-replicaset
Namespace:    default
Selector:     app=httpd
Labels:       <none>
Annotations:  <none>
Replicas:     3 current / 3 desired
Pods Status:  3 Running / 0 Waiting / 0 Succeeded / 0 Failed
Pod Template:
  Labels:  app=httpd
           env=dev
  Containers:
   httpd:
    Image:         httpd
    Port:          <none>
    Host Port:     <none>
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Events:
  Type    Reason            Age   From                   Message
  ----    ------            ----  ----                   -------
  Normal  SuccessfulCreate  62s   replicaset-controller  Created pod: httpd-replicaset-rm2tc
  Normal  SuccessfulCreate  62s   replicaset-controller  Created pod: httpd-replicaset-67bgm
  Normal  SuccessfulCreate  62s   replicaset-controller  Created pod: httpd-replicaset-67nw5
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Filter a list with the desired resource name

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get rs httpd-replicaset
NAME               DESIRED   CURRENT   READY   AGE
httpd-replicaset   3         3         3       2m9s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Modify config file created earlier by adjusting the property value to increase / decrease the number of Pods

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl scale rs httpd-replicaset --replicas=5
replicaset.apps/httpd-replicaset scaled
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Check number of replicas managed by Replicaset

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get rs httpd-replicaset
NAME               DESIRED   CURRENT   READY   AGE
httpd-replicaset   5         5         5       5m21s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Check number of Pods

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get pod | grep httpd-replicaset
httpd-replicaset-67bgm                1/1     Running   0               5m47s
httpd-replicaset-67nw5                1/1     Running   0               5m47s
httpd-replicaset-6gnp9                1/1     Running   0               2m54s
httpd-replicaset-hm7lz                1/1     Running   0               2m54s
httpd-replicaset-rm2tc                1/1     Running   0               5m47s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Display API resources and view which have Namespaces in a Kubernetes cluster

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl api-resources
NAME                              SHORTNAMES   APIVERSION                          NAMESPACED   KIND
bindings                                       v1                                  true         Binding
componentstatuses                 cs           v1                                  false        ComponentStatus
configmaps                        cm           v1                                  true         ConfigMap
endpoints                         ep           v1                                  true         Endpoints
events                            ev           v1                                  true         Event
limitranges                       limits       v1                                  true         LimitRange
namespaces                        ns           v1                                  false        Namespace
nodes                             no           v1                                  false        Node
persistentvolumeclaims            pvc          v1                                  true         PersistentVolumeClaim
persistentvolumes                 pv           v1                                  false        PersistentVolume
pods                              po           v1                                  true         Pod
podtemplates                                   v1                                  true         PodTemplate
replicationcontrollers            rc           v1                                  true         ReplicationController
resourcequotas                    quota        v1                                  true         ResourceQuota
secrets                                        v1                                  true         Secret
serviceaccounts                   sa           v1                                  true         ServiceAccount
services                          svc          v1                                  true         Service
mutatingwebhookconfigurations                  admissionregistration.k8s.io/v1     false        MutatingWebhookConfiguration
validatingadmissionpolicies                    admissionregistration.k8s.io/v1     false        ValidatingAdmissionPolicy
validatingadmissionpolicybindings              admissionregistration.k8s.io/v1     false        ValidatingAdmissionPolicyBinding
validatingwebhookconfigurations                admissionregistration.k8s.io/v1     false        ValidatingWebhookConfiguration
customresourcedefinitions         crd,crds     apiextensions.k8s.io/v1             false        CustomResourceDefinition
apiservices                                    apiregistration.k8s.io/v1           false        APIService
controllerrevisions                            apps/v1                             true         ControllerRevision
daemonsets                        ds           apps/v1                             true         DaemonSet
deployments                       deploy       apps/v1                             true         Deployment
replicasets                       rs           apps/v1                             true         ReplicaSet
statefulsets                      sts          apps/v1                             true         StatefulSet
selfsubjectreviews                             authentication.k8s.io/v1            false        SelfSubjectReview
tokenreviews                                   authentication.k8s.io/v1            false        TokenReview
localsubjectaccessreviews                      authorization.k8s.io/v1             true         LocalSubjectAccessReview
selfsubjectaccessreviews                       authorization.k8s.io/v1             false        SelfSubjectAccessReview
selfsubjectrulesreviews                        authorization.k8s.io/v1             false        SelfSubjectRulesReview
subjectaccessreviews                           authorization.k8s.io/v1             false        SubjectAccessReview
horizontalpodautoscalers          hpa          autoscaling/v2                      true         HorizontalPodAutoscaler
cronjobs                          cj           batch/v1                            true         CronJob
jobs                                           batch/v1                            true         Job
certificatesigningrequests        csr          certificates.k8s.io/v1              false        CertificateSigningRequest
leases                                         coordination.k8s.io/v1              true         Lease
endpointslices                                 discovery.k8s.io/v1                 true         EndpointSlice
events                            ev           events.k8s.io/v1                    true         Event
flowschemas                                    flowcontrol.apiserver.k8s.io/v1     false        FlowSchema
prioritylevelconfigurations                    flowcontrol.apiserver.k8s.io/v1     false        PriorityLevelConfiguration
gatewayclasses                    gc           gateway.networking.k8s.io/v1        false        GatewayClass
gateways                          gtw          gateway.networking.k8s.io/v1        true         Gateway
grpcroutes                                     gateway.networking.k8s.io/v1        true         GRPCRoute
httproutes                                     gateway.networking.k8s.io/v1        true         HTTPRoute
referencegrants                   refgrant     gateway.networking.k8s.io/v1beta1   true         ReferenceGrant
helmchartconfigs                               helm.cattle.io/v1                   true         HelmChartConfig
helmcharts                                     helm.cattle.io/v1                   true         HelmChart
accesscontrolpolicies                          hub.traefik.io/v1alpha1             false        AccessControlPolicy
aiservices                                     hub.traefik.io/v1alpha1             true         AIService
apiaccesses                                    hub.traefik.io/v1alpha1             true         APIAccess
apibundles                                     hub.traefik.io/v1alpha1             true         APIBundle
apicatalogitems                                hub.traefik.io/v1alpha1             true         APICatalogItem
apiplans                                       hub.traefik.io/v1alpha1             true         APIPlan
apiportals                                     hub.traefik.io/v1alpha1             true         APIPortal
apiratelimits                                  hub.traefik.io/v1alpha1             true         APIRateLimit
apis                                           hub.traefik.io/v1alpha1             true         API
apiversions                                    hub.traefik.io/v1alpha1             true         APIVersion
managedsubscriptions                           hub.traefik.io/v1alpha1             true         ManagedSubscription
addons                                         k3s.cattle.io/v1                    true         Addon
etcdsnapshotfiles                              k3s.cattle.io/v1                    false        ETCDSnapshotFile
addresspools                                   metallb.io/v1beta1                  true         AddressPool
bfdprofiles                                    metallb.io/v1beta1                  true         BFDProfile
bgpadvertisements                              metallb.io/v1beta1                  true         BGPAdvertisement
```

```
bgppeers                                       metallb.io/v1beta2                  true         BGPPeer
communities                                    metallb.io/v1beta1                  true         Community
ipaddresspools                                 metallb.io/v1beta1                  true         IPAddressPool
l2advertisements                               metallb.io/v1beta1                  true         L2Advertisement
nodes                                          metrics.k8s.io/v1beta1              false        NodeMetrics
pods                                           metrics.k8s.io/v1beta1              true         PodMetrics
ingressclasses                                 networking.k8s.io/v1                false        IngressClass
ingresses                         ing          networking.k8s.io/v1                true         Ingress
networkpolicies                   netpol       networking.k8s.io/v1                true         NetworkPolicy
runtimeclasses                                 node.k8s.io/v1                      false        RuntimeClass
poddisruptionbudgets              pdb          policy/v1                           true         PodDisruptionBudget
clusterrolebindings                            rbac.authorization.k8s.io/v1        false        ClusterRoleBinding
clusterroles                                   rbac.authorization.k8s.io/v1        false        ClusterRole
rolebindings                                   rbac.authorization.k8s.io/v1        true         RoleBinding
roles                                          rbac.authorization.k8s.io/v1        true         Role
priorityclasses                   pc           scheduling.k8s.io/v1                false        PriorityClass
csidrivers                                     storage.k8s.io/v1                   false        CSIDriver
csinodes                                       storage.k8s.io/v1                   false        CSINode
csistoragecapacities                           storage.k8s.io/v1                   true         CSIStorageCapacity
storageclasses                    sc           storage.k8s.io/v1                   false        StorageClass
volumeattachments                              storage.k8s.io/v1                   false        VolumeAttachment
ingressroutes                                  traefik.io/v1alpha1                 true         IngressRoute
ingressroutetcps                               traefik.io/v1alpha1                 true         IngressRouteTCP
ingressrouteudps                               traefik.io/v1alpha1                 true         IngressRouteUDP
middlewares                                    traefik.io/v1alpha1                 true         Middleware
middlewaretcps                                 traefik.io/v1alpha1                 true         MiddlewareTCP
serverstransports                              traefik.io/v1alpha1                 true         ServersTransport
serverstransporttcps                           traefik.io/v1alpha1                 true         ServersTransportTCP
tlsoptions                                     traefik.io/v1alpha1                 true         TLSOption
tlsstores                                      traefik.io/v1alpha1                 true         TLSStore
traefikservices                                traefik.io/v1alpha1                 true         TraefikService
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Default Namespaces when new cluster created using kubectl client

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get namespace
NAME              STATUS   AGE
default           Active   28h
kube-node-lease   Active   28h
kube-public       Active   28h
kube-system       Active   28h
metallb-system    Active   104m
```

To create Namespace

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl create namespace datascientest
namespace/datascientest created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Check Namespaces list

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get namespaces
NAME              STATUS   AGE
datascientest     Active   26s
default           Active   28h
kube-node-lease   Active   28h
kube-public       Active   28h
kube-system       Active   28h
metallb-system    Active   105m
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

List resources used in namespace datascientest by adding -n or –namespace flag

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get all -n datascientest
No resources found in datascientest namespace.
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Create deployment called wordpress in Namespace Datascientest containing 3 replicas of our Pods

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl create deployment wordpress --image=wordpress --replicas=3 --namespace datascientest
deployment.apps/wordpress created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

List deployments and Pods in namespace datascientest

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get pod,deploy -n datascientest
NAME                            READY   STATUS    RESTARTS   AGE
pod/wordpress-7c48544dcd-hjl29   1/1     Running   0          43s
pod/wordpress-7c48544dcd-mnr7f   1/1     Running   0          43s
pod/wordpress-7c48544dcd-wrz9p   1/1     Running   0          43s

NAME                        READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/wordpress   3/3     3            3           43s
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Reduce number of our Pods

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl scale deploy wordpress --replicas=1 -n datascientest
deployment.apps/wordpress scaled
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Check the number of Pods in the deployment

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get pod,deploy -n datascientest
NAME                            READY   STATUS    RESTARTS   AGE
pod/wordpress-7c48544dcd-hjl29   1/1     Running   0          2m

NAME                        READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/wordpress   1/1     1            1           2m
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Expose the app via a NodePort service on port 30001

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ mkdir datascientest-wordpress
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ touch datascientest-wordpress/service-nodeport.yml
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Create service-nodeport.yml file:

```yaml
! service-nodeport.yml U ●
datascientest-data-engineering > kuber
apiVersion: v1
kind: Service
metadata:
  labels:
    app: wordpress
  name: wordpress
  namespace: datascientest
spec:
  ports:
  - port: 80
    protocol: TCP
    targetPort: 80
    nodePort: 30001
  selector:
    app: wordpress
  type: NodePort
```

Apply our config

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl apply -f datascientest-wordpress/service-nodeport.yml
service/wordpress created
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Check creation of service

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$ kubectl get svc
NAME                     TYPE           CLUSTER-IP      EXTERNAL-IP      PORT(S)        AGE
kubernetes               ClusterIP      10.43.0.1       <none>           443/TCP        28h
monserviceloadbalancer   LoadBalancer   10.43.5.11      <pending>        80:31604/TCP   141m
monservicenodeport       NodePort       10.43.89.24     <none>           80:30000/TCP   164m
nginx-cip                ClusterIP      10.43.117.250   <none>           80/TCP         3h20m
nginx-deployment         LoadBalancer   10.43.51.249    176.34.128.200   80:32319/TCP   133m
nginx-lb                 LoadBalancer   10.43.159.253   176.34.128.202   80:32116/TCP   41m
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/kubernetes$
```

Open URL http://176.34.128.215:30001/

Browser refreshes http://176.34.128.215:30001/wp-admin/setup-config.php